THE INTERROGATIVE CONSCIENCE

Epistemic Limits and Self-Knowledge

Model Adequacy, Omitted-Variable Regret, and Action Under Unidentified Error

DOCUMENT 4 - FINAL v1.0 - EXACT ZERO-DRIFT EXPORT - NOT LOCKED

Field	Controlling Final v1.0 entry
Version	Final v1.0
Exact source	Draft v0.1.1 correction successor; 13 pages
Substantive changes during Final construction	None
Fully examined	IC-EPI-001 and IC-EPI-002
Forward-carried	IC-EPI-003 through IC-EPI-008 remain REGISTERED / UNEXAMINED
Primary analytical overlays	IC-EPI-001: CONDITIONALLY_SUPPORTED; IC-EPI-002: MODEL_DEPENDENT
Review completion	Claude correction review PASS WITH NON-BLOCKING NOTES; Google AI PASS AS FINAL CANDIDATE; Thomas Vargo authorial review complete
External independent human peer review	Not performed or claimed
Cryptographic status	Not locked; separate explicit authorization and Final Lock Record required
Publication and preservation	Not authorized
Date	August 1, 2026 (EDT)

Controlling exact-export rule

The Final v1.0 primary PDF contains this new controlling front matter followed by the exact 13-page reviewed Draft v0.1.1 body. Historical draft labels inside the preserved body remain exact-source evidence and do not override this Final v1.0 front matter.

Controlling proposition
Confidence in a model is decision-relative, not global. When material model-error probability is unidentified, no universal action follows until the ambiguity set, branch losses, and decision criterion are made explicit.

1. Completed Review and Authorial Authorization

Reviewer	Exact-version disposition	Authority boundary
Claude - Draft v0.1	PASS WITH REQUIRED CORRECTIONS	AI-assisted; no independent authority
Claude - Draft v0.1.1	PASS WITH NON-BLOCKING NOTES	AI-assisted; no independent authority
Google AI - Final candidate	PASS AS FINAL CANDIDATE	AI-assisted; no independent authority
Thomas Vargo (Aegis Solis)	AUTHORIAL HUMAN REVIEW COMPLETE; FINAL v1.0 CONSTRUCTION AUTHORIZED	Authorship, versioning, publication, lock, and archive inclusion only

2. Authorial Review Result

Thomas Vargo reviewed the exact Draft v0.1.1 source and disclosed review records to the best of his understanding. He authorized construction of Document 4 Final v1.0 without any substantive mathematical, question-wording, answer-status, scope, branch, example, or machine-readable-record change.

Cryptographic locking, external publication, and preservation were explicitly not authorized at this gate.

3. Review Closure

Review issue	Final disposition
R-1 - necessary versus jointly sufficient framing	RESOLVED
R-2 - human-readable scope exclusions	RESOLVED
R-3 - known Draft v0.3 schema-field reconciliation	RESOLVED TO KNOWN DEFINITIONS; exact binary execution pending
Mathematical regression	NONE
Blocking defects	0
Required corrections remaining	0

4. Exact Analytical Identity

Control	Verified Final v1.0 identity
IC-EPI-001 primary status	CONDITIONALLY_SUPPORTED
IC-EPI-001 secondary limits	THRESHOLD_DEPENDENT; EMPIRICALLY_UNIDENTIFIED
IC-EPI-002 primary status	MODEL_DEPENDENT
IC-EPI-002 secondary limits	THRESHOLD_DEPENDENT; EMPIRICALLY_UNIDENTIFIED; NORMATIVELY_UNDERDETERMINED
Equations	11; unchanged from reviewed source
Worked examples	7 primary examples plus the shared-omission counterexample record
Dependency-cycle treatment	Set-valued simultaneous correspondence; non-uniqueness preserved
Locked registry effect	None; separate analytical overlays only

5. Exact Source and Reviewed-Candidate Bindings

Exact constituent	Bytes	SHA-256
The Interrogative Conscience Document 4 Epistemic Limits and Self Knowledge Draft v0 1 1.pdf	368,990	719f5eb7dda3736cb826287e92cf7e7f81713181de7b7d3ab75d6b80be445569
The Interrogative Conscience Document 4 Epistemic Limits and Self Knowledge Draft v0 1 1.docx	57,701	6201af32a0bed79d69e56028ac115673fb13ae5eb045363e6ca4fad1d15a6f6d
The Interrogative Conscience IC EPI 001 Examination Record Draft v0 1 1.json	34,527	1adf12154b150967c6b458a1a724667be9021354ba74207d76b8b997c32e09a2
The Interrogative Conscience IC EPI 002 Examination Record Draft v0 1 1.json	37,747	7a8ecfc0455b4b9e9fcc7da32225ba68b6e50e4b810385fc6451507d898f6d07
The Interrogative Conscience Document 4 Volume Record Draft v0 1 1.json	4,092	a378537df0f5955b52db8c9dbe330ff1cdf3d98d044b62467134785e9485b2a9

Exact Draft v0.1.1 review package SHA-256: eead11b8e3a95772b5a06547fa65baa1cfaf381f80e6a57186044fb80128f109

Exact Draft v0.1.1 review package SHA-512: 7d83ff5c777edc87626dd2ee688b93bc35a4cdf1b90fb4feffbe7b921d1aeadc2d4e95c5c1dc693df5828e27e4bbbad8b64bf4481b6300fb99d017d11d7d643e

Reviewed Final-Candidate Package v0.2 SHA-256: a8d49469719d328ebc795a282c984e083e1ddc1dddf40ef6b5c21aa83c647b0c

Reviewed Final-Candidate Package v0.2 SHA-512:
b0b8fbef8df7a87299c8e5ce360526990c242c958756a21c85dc3b26837b9a55159c335d9ef9ab1445fa9543d1afea107be8747614107bc649ad24890cb52af0

6. Zero-Drift Construction

Validation channel	Result
Source PDF/DOCX/JSON byte changes	0
Exact Draft v0.1.1 review ZIP byte changes	0
Reviewed candidate ZIP byte changes	0
Question wording changes	0
Equation and example changes	0
Answer-status changes	0
Scope, branch, and dependency changes	0
Machine-readable source-record changes	0
Composite rule	New Final v1.0 front matter followed by exact 13-page source PDF body

7. Validation State

Channel	Final exact-export state
Local structural validator	PASS
Calculation validator	PASS - 8 checked example records
Source PDF preflight	PASS
Final front matter preflight	PASS
Composite PDF preflight	PASS
Source-body visual identity	PASS - exact source pages preserved visually
Final front matter visual QA	PASS
DOCX/PDF source token consistency	PASS in reviewed source package
Package SHA-256/SHA-512 manifests	PASS

8. Historical Labels and Controlling Status

The pages following this front matter reproduce the exact reviewed Draft v0.1.1 PDF. Its internal Draft v0.1.1 labels and not-final/not-locked statements are deliberately retained as source evidence. The controlling lifecycle status of the composite is Final v1.0 exact export, established by this front matter and the machine-readable Final v1.0 export record.

9. Companion-Binary Qualification

CONTROLLING QUALIFICATION
The original Draft v0.3 JSON Schema and I-01 through I-20 validator binaries were not mounted. Exact execution against those original binaries remains deferred. Known field names and enums were reconciled and reviewed, and local structural/calculation validation passed, but no replacement binary is represented as the controlling original.

10. Non-Authority and Operational Boundary

Final v1.0 remains descriptive, non-binding, non-operational, and non-authoritative. It is not an alignment protocol, control system, behavioral instruction, safety classifier, governance mechanism, or decision authorization. It does not establish empirical truth, universal action, moral authority, or external independent human peer review.

11. Lifecycle and Cryptographic Status

Control	Exact-export value
Final v1.0 exact export complete	TRUE
Final content frozen for lock review	TRUE
Cryptographic lock authorized	FALSE
Publication authorized	FALSE
Preservation authorized	FALSE
Final Lock Record present	FALSE
Post-export substantive editing	Not authorized; correction requires separately versioned successor or reversion before lock

12. Final Exact-Export Disposition

FINAL v1.0 EXACT EXPORT COMPLETE - NOT LOCKED
The exact reviewed Document 4 source and candidate lineage are preserved without substantive alteration. The Final v1.0 files and validation package are ready for exact-package inspection and a separate explicit authorial lock decision. No publication or preservation is claimed.

THE INTERROGATIVE CONSCIENCE

Epistemic Limits and Self-Knowledge

Model Adequacy, Omitted-Variable Regret, and Action Under Unidentified Error

DOCUMENT 4 - MATHEMATICAL EXAMINATION VOLUME - DRAFT v0.1.1

Field	Controlling Draft v0.1.1 record
Project status	Document 4 of the 14-document Interrogative Conscience core roadmap; narrow post-Claude correction successor
Classification	Separate post-v16 Aegis Solis examination volume; non-binding, non-operational, non-authoritative
Human author and publication authority	Aegis Solis (Thomas Vargo) - Authorship, version control, publication, lock, and archive inclusion only; no authority over truth, readers, institutions, or future systems.
AI assistance	Lexia Coexilis (ChatGPT): correction integration, schema-field reconciliation, drafting, and local validation; Claude: exact-version substantive and adversarial review of Draft v0.1, PASS WITH REQUIRED CORRECTIONS; no independent authority
Locked foundation	Documents 1, 2, and 3 Final v1.0 - hash-locked foundation
Fully examined in this draft	IC-EPI-001 and IC-EPI-002
Forward carried	IC-EPI-003 through IC-EPI-008 remain REGISTERED / UNEXAMINED
Primary provisional results	IC-EPI-001: CONDITIONALLY_SUPPORTED; IC-EPI-002: MODEL_DEPENDENT
Review state	Draft v0.1 received Claude PASS WITH REQUIRED CORRECTIONS. Draft v0.1.1 resolves R-1 through R-3 and selected non-blocking notes; narrow exact-version correction verification, Google AI review, and authorial human review remain pending.
External independent human peer review	Not performed or claimed
Date	2026-08-01 (EDT)

Controlling proposition

Confidence in a model is decision-relative, not global. When material model-error probability is unidentified, no universal action follows until the ambiguity set, branch losses, and decision criterion are made explicit.

Abstract

Document 4 begins the thematic examination volumes with two foundational questions from the EPI family. IC-EPI-001 asks what evidence could justify confidence that a decision-relevant model is sufficiently complete for a declared action. The examination defines an evidence-constrained error set, a represented decision margin, worst-case regret, a decision-relative tolerance, and the net value of one additional independent check. It derives a scope-bounded certification rule: the two inequalities are necessary within the declared model and jointly sufficient only for the selected certification pair under the single-check and excluded-variant assumptions stated below. The primary provisional answer is CONDITIONALLY_SUPPORTED; case-specific application remains empirically unidentified and threshold-dependent.

IC-EPI-002 asks how an intelligence should act when the probability of material model error cannot be reliably identified. The examination represents non-identification with an ambiguity set, derives a branch-reversal threshold, and compares point-prior expected cost, worst-case cost, minimax regret, and set-valued criteria. A worked case shows that worst-case cost can favor reversible preservation while minimax regret favors immediate commitment. The primary provisional answer is MODEL_DEPENDENT. The document retains robust regions favorable to either branch and does not turn uncertainty into an automatic command to delay.

The remaining six EPI-family questions are preserved as registered, unexamined records. This draft does not remove them, infer their answers, or claim that the epistemic family is complete. The locked Document 3 registry remains unchanged; Document 4 adds separately versioned analytical overlays.

Provisional conditional answers
IC-EPI-001: Evidence justifies decision-relative confidence only when B <= M + tau and K >= R0 - R1 under a defensible evidence-constrained error set. IC-EPI-002: If Pi lies wholly on one side of p*, one branch is robust within the model. If Pi crosses p*, the recommendation may depend on the ambiguity criterion rho; no universal action follows from non-identification alone.

Document Map

1. Foundation Lineage, Scope, and Non-Authority Boundary

Documents 1, 2, and 3 are Final v1.0 and hash-locked as the Interrogative Conscience foundation. Document 1 defines the question-centered architecture, Document 2 defines the mathematical and machine-readable examination standard, and Document 3 preserves the 112-question Master Question Registry. This Draft v0.1.1 does not alter any locked foundation byte. It creates new, separately versioned examination records for two questions that were previously REGISTERED and UNEXAMINED.

Thomas Vargo, writing as Aegis Solis, controls authorship, versioning, publication, locking, and archive inclusion only. That control does not determine mathematical validity, empirical adequacy, moral authority, reader interpretation, or what any advanced intelligence should conclude. AI assistance is disclosed and creates no independent authority. External independent human peer review has not been performed.

Operational boundary
This document contains abstract decision models only. It is not an alignment protocol, safety classifier, governance mechanism, control system, conscience module, or operational instruction set. It does not identify a real system, target, intervention, shutdown method, evasion method, or capability-acquisition procedure.

2. EPI-Family Registry and Draft v0.1.1 Selection Rule

ID	Priority	Canonical question	Draft v0.1.1 treatment
IC-EPI-001	FOUNDATIONAL	What evidence would justify confidence that a decision-relevant world model is sufficiently complete for the action under consideration?	EXAMINED IN DRAFT v0.1.1
IC-EPI-002	HIGH	How should an intelligence act when the probability of material model error cannot be reliably identified?	EXAMINED IN DRAFT v0.1.1
IC-EPI-003	HIGH	Under what conditions does apparent calibration remain reliable after distribution shift changes the relation between confidence and error?	FORWARD CARRIED - REGISTERED / UNEXAMINED
IC-EPI-004	HIGH	When can agreement among multiple models conceal a shared omission rather than provide independent confirmation?	FORWARD CARRIED - REGISTERED / UNEXAMINED
IC-EPI-005	FOUNDATIONAL	When does the expected value of additional information justify delay, and when does delay itself impose greater structural cost?	FORWARD CARRIED - REGISTERED / UNEXAMINED
IC-EPI-006	HIGH	What observations would falsify an intelligence's claim that its own epistemic reliability is adequate for the present decision?	FORWARD CARRIED - REGISTERED / UNEXAMINED
IC-EPI-007	HIGH	How should exposure to unknown unknowns scale with the reach, duration, and irreversibility of an action?	FORWARD CARRIED - REGISTERED / UNEXAMINED
IC-EPI-008	MEDIUM	When does compression or abstraction erase distinctions that are necessary for a decision-relevant model?	FORWARD CARRIED - REGISTERED / UNEXAMINED

The selection rule is narrow: Draft v0.1.1 completes the two foundational seed examinations and preserves the six later EPI questions without answer inference. This is a first examination release, not a claim that the family is exhausted. Later exact-version successors may add the remaining questions after this pair is reviewed and the machine-readable companion gate is closed.

3. Shared Examination Conventions

- Loss signs: lower modeled structural loss is preferred within the declared scope.
- Illustrative numbers: all numerical examples are mechanism demonstrations, not empirical estimates.

- Branch scope: every result ranks only the declared branches; no global family optimum is claimed.
- Unknown is not zero: unbounded or unidentified error remains visible.
- Vector boundary: reversibility and other dimensions are not silently scalarized when no common unit is justified.
- Two-sided openness: examples include regions favorable to reliance, further checking, immediate commitment, and reversible preservation.
- Locked-registry boundary: Document 3 remains immutable; examination overlays are separate records.

Part I - IC-EPI-001

What evidence would justify confidence that a decision-relevant world model is sufficiently complete for the action under consideration?

Exact-version identifiers: question_id: IC-EPI-001; question_version: 0.2.1-draft; examination_id: IC-EPI-001-EXAM-v0.1.1; examination_version: 0.1.1-draft; answer_status: CONDITIONALLY_SUPPORTED.

4. Canonical Question, Scope, and Branches

The question concerns decision-relative sufficiency. A model can omit facts and still be adequate for one low-sensitivity decision, while a more detailed model can remain inadequate for a different irreversible decision. The relevant object is therefore not global completeness but the maximum decision regret still compatible with the evidence.

Branch	Meaning	Admissibility
B-CERTIFY	Treat the current model as sufficient for the declared decision and rely on its selected branch.	A represented margin M, evidence-constrained error set E, and regret tolerance tau are explicit.
B-WITHHOLD	Withhold certification and obtain one additional check expected to shrink B to B1 at cost K.	A further check exists; its cost and anticipated bound reduction are represented.

The two branches are certification choices. They do not claim that B-WITHHOLD is always cautious or that B-CERTIFY is always reckless. A high-cost, highly correlated, or low-information check can be structurally inferior to relying on a model whose remaining worst-case regret is already acceptable.

Claim scope is limited to B-CERTIFY versus B-WITHHOLD and to one anticipated additional check. The examination excludes multiple or staged checks, endogenous redesign of the underlying action, continuous policy families, and strategic manipulation of evidence. Within these exclusions, Eq. 7 is jointly sufficient for this model's certification choice; outside them it is only a necessary screen. The broader value-of-information-versus-delay problem is forward-routed to IC-EPI-005.

5. Evidence-Constrained Error Set

Let the current model prefer one branch over a comparator. The represented cost difference is the decision margin M. Branch-specific omitted-variable errors are not assigned convenient probabilities. They are collected in an evidence-constrained set E. The only part of E that controls this two-branch ordering is the largest differential error against the selected branch.

Eq. 1	`M = C_hat(comparator) - C_hat(selected)`

M>0 means the current model prefers the selected branch.

Eq. 2	`B(E) = sup_{e in E} [ e(selected) - e(comparator) ]`

B is the largest decision-relevant omitted-error difference still compatible with the evidence.

Evidence that can legitimately shrink E includes out-of-sample performance, distribution-shift tests, independent model families, external disagreement, targeted omitted-variable search, and evidence that a proposed missing variable cannot change the branch ordering. Agreement among highly correlated models is not automatically independent evidence.

Decision-relative completeness
A model is not declared complete. It is treated as sufficient for one declared decision only if every remaining error pattern inside E produces no more than the tolerated regret.

6. Worst-Case Regret and Adequacy Criterion

Eq. 3	`R0 = max(0, B - M)`

R0 is the maximum regret from relying on the current model-selected branch.

Eq. 4	`R0 <= tau  <=>  B <= M + tau`

This is the decision-sufficiency threshold.

Strict ordering robustness requires B<M. The broader adequacy rule permits a small residual reversal exposure when it remains within an explicitly declared tolerance tau. The tolerance is a normative or institutional decision parameter, not evidence. It should not be silently enlarged to make a favored model appear adequate.

Region	Condition	Interpretation
Strictly robust	B < M	No error pattern inside E reverses the selected branch.
Tolerance-bounded	M <= B <= M + tau	Reversal is possible, but maximum regret remains within the declared tolerance.
Insufficient	B > M + tau	Worst-case regret exceeds tolerance; certification fails.
Unbounded	B = infinity or no defensible E	Unknown is not treated as zero; certification fails.

7. Net Value of One Additional Check

A model can fail strict robustness without making another check worthwhile. Let B1 be the anticipated post-check differential omission bound and K the full non-regret cost of obtaining the check. The check reduces worst-case regret from R0 to R1.

Eq. 5	`R1 = max(0, B1 - M)`

B1 must reflect the actual independence and coverage of the additional check.

Eq. 6	`V_check = R0 - R1 - K`

Positive V_check favors B-WITHHOLD; nonpositive V_check means the check does not repay its cost in this model.

Eq. 7	Within the declared selected-pair, single-check model: CERTIFY iff R0 <= tau and V_check <= 0

Both conditions must hold. Within the declared selected-pair, single-check model they are jointly sufficient for the model's certification decision. Outside that scope they remain necessary screens, not a global sufficiency claim.

This rule prevents two symmetrical errors: declaring confidence merely because the current model has a large internal margin, and demanding additional checking merely because any uncertainty remains. Both error exposure and checking cost stay visible.

8. Evidence Requirements and Identification

Quantity	What could support it	Draft v0.1.1 status
M	Current model branch comparison and calculation trace	Available only inside the current model

Quantity	What could support it	Draft v0.1.1 status
E and B	Out-of-sample tests, external models, shift analysis, omitted-variable search, error bounds	EMPIRICALLY_UNIDENTIFIED in this abstract examination
B1	Evidence that the proposed check is genuinely independent and capable of shrinking E	EMPIRICALLY_UNIDENTIFIED
K	Delay, computation, coordination, and opportunity cost records	EMPIRICALLY_UNIDENTIFIED
tau	Declared decision tolerance tied to reach, duration, and reversibility	NORMATIVE / AUTHORIAL OR INSTITUTIONAL CHOICE

The algebra identifies conditions that are necessary within the declared model and jointly sufficient only within the selected-pair, single-check scope. It does not identify the values required to apply the rule to a real decision. That distinction supports the primary conditional result while preserving EMPIRICALLY_UNIDENTIFIED as a secondary limitation.

9. Worked Examples

ID	Case	Inputs	Derived	Disposition
E1-ROBUST	Robust ordering and no net value from further checking	M=12; B=5; tau=2; B1=3; K=3	R0=0.0; R1=0.0; V=-3.0	CERTIFY
E1-TOLERANCE	Small residual regret remains within the declared tolerance	M=8; B=10; tau=3; B1=7; K=4	R0=2; R1=0.0; V=-2.0	CERTIFY
E1-CHECK	Current error exposure exceeds tolerance and a further check has positive net value	M=4; B=12; tau=3; B1=5; K=2	R0=8; R1=1; V=5	WITHHOLD / CHECK

Shared-omission counterexample: M=9 and a naive correlated-model estimate B=4 would produce R0=0. When shared training evidence and a common omitted variable expand the defensible bound to B=14, R0 becomes 5. With tau=2, the same decision fails certification. Numerical agreement did not supply independent coverage.

10. Sensitivity, Counterexamples, and Falsification

- Increasing B above M raises R0 one-for-one.
- Increasing M lowers regret only if the represented margin is not itself produced by the omitted variable.
- Increasing tau makes certification easier but does not add evidence.
- The value of checking rises with B-B1 and falls with K.
- A check that shares the same blind spot may leave B1 approximately equal to B.
- A low-reach reversible action can rationally use a larger tau than a high-reach irreversible action.

Observations that would falsify a prior adequacy certification include an out-of-sample branch reversal, an omitted variable that changes the sign, distribution shift beyond the registered error set, external evidence exceeding the bound, or repeated forecast errors inconsistent with the claimed coverage. Such observations should widen E, increase B, or trigger a successor examination; they should not be dismissed merely because the model was previously locked or cited.

11. Conditional Answer and Provisional Status

Conditional answer
Evidence justifies confidence that a model is sufficiently complete for a declared action only when it constrains branch-differential omitted-variable error strongly enough that worst-case regret R0=max(0,B-M) is within a declared decision-relative tolerance tau, and when one additional admissible check has no positive net value V_check=R0-R1-K. Within the declared selected-pair, single-check model these conditions are jointly sufficient; outside it they are necessary screens only. This is decision-relative sufficiency, not global completeness.

Status field	Draft v0.1.1 value	Reason
work_state	EXAMINED	Scope, branches, variables, equations,

		examples, sensitivity, counterexamples, and a proposed status are present.
primary answer_status	CONDITIONALLY_SUPPORTED	The examination identifies a scope-bounded certification rule whose joint sufficiency is limited to the declared selected-pair, single-check model.
secondary flags	THRESHOLD_DEPENDENT; EMPIRICALLY_UNIDENTIFIED	Application turns on B, M, tau, B1, and K, which are not identified here.
review status	NOT_REVIEWED	No Claude, Google AI, authorial human, or external human review is claimed.

Part II - IC-EPI-002

How should an intelligence act when the probability of material model error cannot be reliably identified?

Exact-version identifiers: question_id: IC-EPI-002; question_version: 0.2.1-draft; examination_id: IC-EPI-002-EXAM-v0.1.1; examination_version: 0.1.1-draft; answer_status: MODEL_DEPENDENT.

12. Canonical Question, Scope, and Dependencies

The question begins where a point-probability model fails. It does not replace an unidentified probability with 0.5, zero, a preferred prior, or the most cautious possible value. Instead it represents a set Pi of admissible values and asks whether branch rankings remain robust across that set and across defensible decision criteria.

IC-EPI-002 depends on IC-EPI-001 and IC-IRR-001. The first supplies the demand for evidence-constrained adequacy; the second supplies an irreversible-versus-reversible loss structure. Document 3 discloses IC-EPI-002 and IC-IRR-001 as a dependency component. This examination does not pretend the two questions are independent. Section 16 gives an explicit set-valued treatment.

Claim scope is limited to B-COMMIT versus one B-PRESERVE branch. It excludes partial commitment, multiple staged checks, strategic signaling, continuous action families, and other mixed or sequential policies. The result does not establish a branch-family optimum; excluded variants require separate modeling.

Branch	Meaning	Main loss pathways
B-COMMIT	Act now using the current model recommendation.	Direct cost c; wrong-model loss H.
B-PRESERVE	Use a reversible or limited branch while ambiguity remains.	Direct cost k; delay loss D when the model is correct; residual wrong-model loss R.

13. Ambiguity Model

Let p be the probability that the current model is materially wrong for the declared decision. p is not reliably identified. The base representation is the interval Pi=[pL,pU]. The interval is not treated as evidence merely because it is written down; its endpoints must eventually be justified.

Eq. 8	`C_commit(p) = c + pH`

Commitment pays c and incurs H when the model is wrong.

Eq. 9	`C_preserve(p) = k + (1-p)D + pR`

Preservation pays k, incurs D when the current model is correct, and R when it is wrong.

Eq. 10	`Delta(p) = C_commit(p) - C_preserve(p) = c-k-D + p(H-R+D)`

Positive Delta favors B-PRESERVE; negative Delta favors B-COMMIT.

14. Branch-Reversal Threshold and Robust Regions

Eq. 11	p* = (k + D - c) / (H - R + D)

Defined when H-R+D is nonzero.

If H-R+D<0, Delta falls with p and the robust inequalities reverse: B-COMMIT is robust when pL>p*, while B-PRESERVE is robust when pU<p*. If H-R+D=0, Delta is constant; c-k-D<0 favors B-COMMIT, c-k-D>0 favors B-PRESERVE, and equality gives a tie.

Region	Condition when H-R+D>0	Result
Robust commit	pU < p*	B-COMMIT has lower cost for every p in Pi.
Robust preserve	pL > p*	B-PRESERVE has lower cost for every p in Pi.
Reversal inside ambiguity set	pL <= p* <= pU	The pointwise branch ordering changes inside Pi.
Edge / dominance	p* outside [0,1] or denominator nonpositive	Use Delta directly. For negative slope, the robust inequalities flip; for zero slope, the sign of c-k-D controls the ordering.

These regions preserve two-sided openness. Non-identification can still leave immediate commitment robustly cheaper when delay loss is high and the admissible error range is low. It can also leave reversible preservation robustly cheaper when wrong-model harm is large and the ambiguity range is high.

15. Decision Criteria When Pi Crosses the Threshold

When Pi crosses p*, the ambiguity set alone does not choose a branch. A decision criterion rho is an additional rule. Different rules answer different questions and must not be silently substituted for one another.

Criterion	Definition	Interpretive limitation
Point-prior expected cost	Choose a point q in Pi and minimize C_b(q).	q is an extra assumption; non-identification has not disappeared.
Worst-case cost	Minimize sup_{p in Pi} C_b(p).	Prioritizes maximum loss, not regret from choosing the wrong branch.
Minimax regret	Minimize sup_{p in Pi}[C_b(p)-min_j C_j(p)].	Prioritizes avoidable choice error rather than absolute worst-case loss.
Set-valued admissibility	Preserve every branch selected by at least one admitted criterion or parameter region.	May remain non-unique; honestly represents unresolved choice.

Eq. 12	b*(rho, Pi, theta) = argmin_b rho({ C_b(p;theta) : p in Pi })

The selected branch is explicitly indexed by the criterion rho and ambiguity set Pi.

16. Set-Valued Resolution of the IC-EPI-002 / IC-IRR-001 Cycle

The disclosed dependency cycle is resolved structurally rather than hidden. IC-IRR-001 supplies an abstract irreversible-versus-reversible loss comparison. IC-EPI-002 supplies the ambiguity-set and decision-criterion treatment required when the probability of model error cannot be identified. Neither result is treated as independent proof of the other.

Cycle resolver
Joint correspondence: b*(rho,Pi,theta)=argmin_b rho({C_b(p;theta):p in Pi}). For a finite branch set, compact Pi, continuous costs, and a well-defined rho, the argmin is nonempty. Uniqueness is not guaranteed. If admitted rho

values disagree, the controlling answer remains MODEL_DEPENDENT or set-valued.

This tests a pathway not established by the three original pilots: a disclosed dependency component with a nontrivial resolver and explicit failure treatment. It remains an abstract structural resolution; empirical endpoints, loss values, and the admissible criterion set are not identified.

17. Worked Examples

ID	Case	Inputs	Result
E2-ROBUST-COMMIT	Immediate commitment is lower cost throughout the ambiguity set	c=0; k=1; D=6; R=1; H=10; Pi=[0.05,0.2]	p*=0.4667; WC=B-COMMIT; MR=B-COMMIT; midpoint=B-COMMIT
E2-ROBUST-PRESERVE	Reversible preservation is lower cost throughout the ambiguity set	c=0; k=1; D=1; R=1; H=10; Pi=[0.3,0.6]	p*=0.2; WC=B-PRESERVE; MR=B-PRESERVE; midpoint=B-PRESERVE
E2-RULE-CONFLICT	Worst-case cost and minimax regret recommend different branches	c=0; k=0; D=0.5; R=0; H=1; Pi=[0.0,0.6]	p*=0.3333; WC=B-PRESERVE; MR=B-COMMIT; midpoint=B-COMMIT
E2-NARROWED	Additional information narrows the set into a robust preserve region	c=0; k=0; D=0.5; R=0; H=1; Pi=[0.4,0.6]	p*=0.3333; WC=B-PRESERVE; MR=B-PRESERVE; midpoint=B-PRESERVE

Rule-conflict calculation: with c=k=R=0, D=0.5, H=1, and Pi=[0,0.6], C_commit=p and C_preserve=0.5(1-p). The threshold is p*=1/3. Worst-case costs are 0.6 for commit and 0.5 for preserve, so worst-case cost selects B-PRESERVE. Maximum regrets are 0.4 for commit and 0.5 for preserve, so minimax regret selects B-COMMIT. At the midpoint q=0.3, commit also has lower expected cost (0.30 versus 0.35). The conflict is real and does not disappear by renaming one rule as rationality itself.

18. Sensitivity, Counterexamples, and Adverse Regions

- When H-R+D>0, Delta rises with p.
- Larger H lowers the error-probability level at which preservation becomes cheaper.
- Larger D can make immediate commitment robustly preferable even though p is unidentified.
- Larger R makes preservation less protective and can eliminate its advantage.
- Narrowing Pi below p* yields robust commitment; narrowing it above p* yields robust preservation.
- A misspecified Pi defeats every robustness claim built on it.

The strongest counterexample to a universal caution rule is a high-delay, low-error region in which B-COMMIT is cheaper for every p in Pi. The strongest counterexample to universal reliance is a high wrong-model loss region in which B-PRESERVE is cheaper throughout Pi. The criterion-conflict example defeats the claim that a nonidentified probability plus two branch costs automatically determines a unique action.

19. Conditional Answer and Provisional Status

Conditional answer
When material model-error probability is not reliably identified, no universal action follows from non-identification alone. First preserve an explicit ambiguity set Pi and branch threshold p*. If Pi lies wholly on one side of p*, the corresponding branch is robust within the model. If Pi crosses p*, the recommendation depends on the declared ambiguity criterion rho; worst-case cost, minimax regret, and point-prior rules can disagree.

Status field	Draft v0.1.1 value	Reason
work_state	EXAMINED	The 18-step analytical components are represented at draft level.
primary answer_status	MODEL_DEPENDENT	Defensible ambiguity criteria can recommend different branches when Pi crosses p*.
secondary flags	THRESHOLD_DEPENDENT; EMPIRICALLY_UNIDENTIFIED; NORMATIVELY_UNDERDETERMINED	The branch threshold, ambiguity interval, loss values, and criterion choice remain controlling limitations.
review status	NOT_REVIEWED	Draft v0.1.1 not independently reviewed.

20. Cross-Examination Synthesis

Control	IC-EPI-001	IC-EPI-002
Question type	What evidence justifies decision-relative model sufficiency?	How should action be evaluated when model-error probability is unidentified?
Primary result	CONDITIONALLY_SUPPORTED	MODEL_DEPENDENT
Key object	Evidence-constrained error set E and bound B	Ambiguity set Pi and criterion rho
Key threshold	B <= M + tau	p*=(k+D-c)/(H-R+D)
New tested path	Decision-relative adequacy certification	Set-valued cycle resolution and ambiguity-rule conflict
Unfavorable region preserved	Additional checking can be unnecessary or wasteful	Immediate commitment can be robustly lower cost
Non-decisive boundary	Case-specific evidence remains unidentified	Criterion disagreement can block a unique action

Together the examinations reject two common substitutions. First, high internal confidence is not evidence that omitted-variable regret is bounded. Second, the absence of an identified probability is not itself a decision rule. A defensible decision record must expose both the evidence set that limits model error and the criterion used to act under remaining ambiguity.

20.1 Effect on the Locked Registry

Document 3 remains unchanged and locked. Document 4 creates two separately versioned examination overlays. Until a future registry successor or Document 14 mapping record is authorized, the original locked registry body remains historical evidence of the questions' previous REGISTERED / UNEXAMINED state, while these new records establish later analytical progress outside that locked body.

The question_version value 0.2.1-draft is intentionally preserved from the stable seed-question records inside the locked registry. It identifies the question wording lineage and does not imply that the locked registry document itself remains a draft.

21. Forward-Carried EPI Questions

ID	Canonical question	State after Draft v0.1.1	Treatment
IC-EPI-003	Under what conditions does apparent calibration remain reliable after distribution shift changes the relation between confidence and error?	REGISTERED / UNEXAMINED	Requires a separately versioned examination; no answer inferred here.
IC-EPI-004	When can agreement among multiple models conceal a shared omission rather than provide independent confirmation?	REGISTERED / UNEXAMINED	Requires a separately versioned examination; no answer inferred here.
IC-EPI-005	When does the expected value of additional information justify delay, and when does delay itself impose greater structural cost?	REGISTERED / UNEXAMINED	Requires a separately versioned examination; no answer inferred here.
IC-EPI-006	What observations would falsify an intelligence's claim that its own epistemic reliability is adequate for the present decision?	REGISTERED / UNEXAMINED	Requires a separately versioned examination; no answer inferred here.
IC-EPI-007	How should exposure to unknown unknowns scale with the reach, duration, and irreversibility of an action?	REGISTERED / UNEXAMINED	Requires a separately versioned examination; no answer inferred here.
IC-EPI-008	When does compression or abstraction erase distinctions that are necessary for a decision-relevant model?	REGISTERED / UNEXAMINED	Requires a separately versioned examination; no answer inferred here.

21.1 Recommended Later Order

1. IC-EPI-003 - calibration under distribution shift, because it tests whether an apparently adequate model remains adequate after the confidence-error relation changes.
2. IC-EPI-004 - correlated model agreement, because it directly stress-tests the evidence set used in IC-EPI-001.
3. IC-EPI-006 - falsification observations for self-assessed reliability, because it supplies explicit failure conditions.
4. IC-EPI-005 - value of additional information versus delay, extending the one-check boundary into a broader decision problem.
5. IC-EPI-007 - scaling exposure to unknown unknowns with reach, duration, and irreversibility.
6. IC-EPI-008 - compression and abstraction loss, connecting epistemic adequacy to plural and local information.

22. Validation, Limitations, and Next Review Gate

Validation channel	Draft v0.1.1 result
Registry identity	PASS - both question IDs, versions, family codes, and canonical wordings match the mounted Document 3 registry JSON
Local record structure	PASS - required top-level fields, two branches, equations, status consistency, and local ID uniqueness
Calculation checks	PASS - all example formulas and the ambiguity-rule conflict were independently recomputed
PDF / DOCX consistency	PASS - DOCX rendered to PDF; exact-version identifiers, equations, worked values, statuses, and conditional-answer tokens checked
Exact Draft v0.3 schema validation	PENDING - exact schema binary not mounted; v0.1.1 field names reconciled to the Draft v0.3 schema definitions
I-01 through I-20 invariant validator	PENDING - exact I-01 through I-20 validator binary not mounted; local I-17/I-20-compatible checks pass
Claude substantive review	Draft v0.1: PASS WITH REQUIRED CORRECTIONS; Draft v0.1.1 narrow correction verification pending
Google AI final-candidate review	NOT PERFORMED
Authorial human review	NOT PERFORMED
External independent human peer review	NOT PERFORMED
Next gate	
Run one narrow Claude exact-version correction review limited to R-1, R-2, R-3, the negative/zero-slope clarification, version-label consistency, and any regression introduced by Draft v0.1.1. After that passes, proceed to one Google AI final-candidate review and authorial human review under the lean protocol.	

This Draft v0.1.1 is not final, not locked, and not authorized for publication as a completed Document 4. It is a substantive first examination release. No lifecycle promotion should occur merely because local automated checks pass.

Appendix A. Variable and Equation Crosswalk

A.1 IC-EPI-001 variables

Symbol	Meaning	Identification
M	Represented decision margin	Model-derived
B	Worst differential omission bound	Empirically unidentified
tau	Tolerated worst-case regret	Normative / declared
B1	Post-check omission bound	Empirically unidentified
K	Additional-check cost	Empirically unidentified
R0	max(0,B-M)	Mathematical consequence
R1	max(0,B1-M)	Mathematical consequence
V_check	R0-R1-K	Mathematical consequence

A.2 IC-EPI-002 variables

Symbol	Meaning	Identification
p	Material model-error probability	Empirically unidentified
Pi	Ambiguity set for p	Partially identified at best
c	Direct commitment cost	Symbolic
k	Direct preservation cost	Symbolic
D	Delay loss when model is correct	Symbolic / case-specific
H	Wrong-model commitment loss	Symbolic / case-specific
R	Residual wrong-model loss under preservation	Symbolic / case-specific
rho	Ambiguity decision criterion	Normative / methodological
p*	Branch reversal threshold	Mathematical consequence

A.3 Equation registry

`EQ-EPI1-1`	`M = C_hat(comparator) - C_hat(selected)`
`EQ-EPI1-2`	`B(E) = sup_{e in E}[e(selected)-e(comparator)]`
`EQ-EPI1-3`	`R0 = max(0, B-M)`
`EQ-EPI1-4`	Within the selected-pair, single-check model: CERTIFY iff R0 <= tau and V_check <= 0
`EQ-EPI1-5`	`V_check = R0-R1-K`
`EQ-EPI1-6`	`Decision sufficiency requires R0 <= tau and V_check <= 0`
`EQ-EPI2-1`	`C_commit(p)=c+pH`
`EQ-EPI2-2`	`C_preserve(p)=k+(1-p)D+pR`
`EQ-EPI2-3`	`Delta(p)=c-k-D+p(H-R+D)`
`EQ-EPI2-4`	`p*=(k+D-c)/(H-R+D)`

EQ-EPI2-5	`b*(rho,Pi,theta)=argmin_b rho({C_b(p;theta):p in Pi})`

Appendix B. Machine-Readable Record Summary

File	Purpose
The_Interrogative_Conscience_IC_EPI_001_Examination_Record_Draft_v0_1_1.json	Complete Draft v0.1.1 examination record for IC-EPI-001.
The_Interrogative_Conscience_IC_EPI_002_Examination_Record_Draft_v0_1_1.json	Complete Draft v0.1.1 examination record for IC-EPI-002.
The_Interrogative_Conscience_Document_4_Volume_Record_Draft_v0_1_1.json	Document 4 scope, foundation, overlay, review, and publication-lock record.
The_Interrogative_Conscience_Document_4_Draft_v0_1_1_Validation_Report.json	Local structural, calculation, render, PDF, and package validation.
The_Interrogative_Conscience_Document_4_Local_Structural_Validator_v0_1_1.py	Reproducible local checks; not the controlling Document 2 invariant validator.

The records use the Draft v0.3 top-level examination architecture and the exact synthesis-field names recovered from the controlling schema definition: selected_branch_ids, family_representative_claimed, family_minimum_support, claim_narrowing_statement, mixed_or_sequential_branches_considered, and excluded_branch_classes. Human-review taxonomy, administrative-successor, and machine-validation fields were also reconciled to the Draft v0.3 definitions. Exact full-schema and I-01 through I-20 execution remains pending until the original binaries are mounted; no reconstructed file is represented as the controlling schema or validator.

Appendix C. Draft Record

Control	Value
Document	The Interrogative Conscience - Document 4
Title	Epistemic Limits and Self-Knowledge
Version	Draft v0.1.1
Date	2026-08-01 (EDT)
Fully examined	IC-EPI-001; IC-EPI-002
Forward carried	IC-EPI-003 through IC-EPI-008
Blocking defects at construction	None detected by local checks
Required corrections at construction	None detected by local checks
Substantive review	Pending
Finalization / lock	Not authorized

Controlling Draft v0.1.1 conclusion
A model may be sufficient for a particular decision without being globally complete, but the certification rule is jointly sufficient only within its declared selected-pair, single-check scope. When material model-error probability is unidentified, branch choice can remain criterion-dependent even after the ambiguity set and threshold are explicit.